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OCEAN GUARD AI

An AI-powered system for marine protection and research

NAMING & DEFINITIONS

Underwater Drones → vehicles that capture video, sonar, and environmental data underwater.

Sonar Arrays → Sensor networks that use sound waves to detect and track marine animal movements.

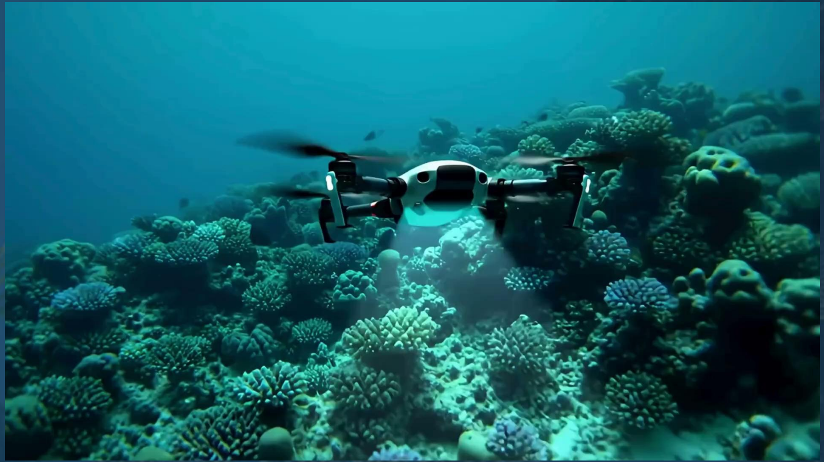
AI Models → Trained algorithms that identify species and classify behaviors in real time.

Safe Deterrents → Non-harmful methods (sound pulses, light signals) to redirect threats without harming animals.

PROJECT OVERVIEW

OceanGuard AI is a large-scale monitoring system for large marine animals.

It uses underwater drones, sonar and AI placed in parts of the ocean and detects threats to various marine species in real time.



Scientists can activate deterrents like sound waves or collect data for studies.

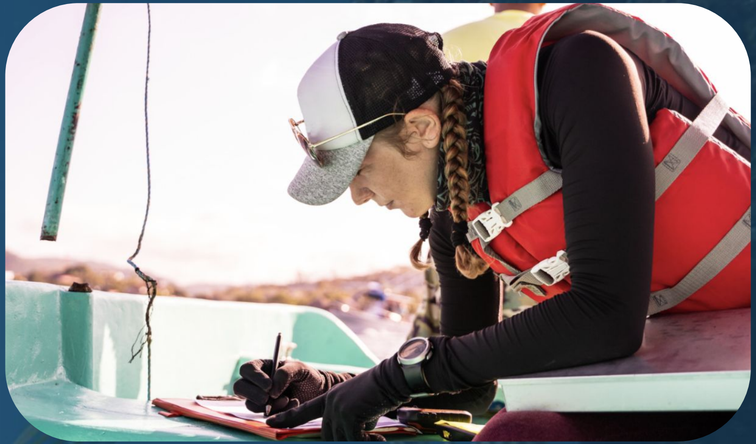
PURPOSE & GOALS

Marine species and ecosystems are increasingly threatened by pollution, human activity, and climate change.

OceanGuard AI will support scientists protect species, fight pollution, and gather data on ocean ecosystems.

USERS & STAKEHOLDERS

- Marine biologists & conservationists
- Governments & NGOs
- Aquaculture and fisheries
- Every day, ocean lovers



IMPACT

- Protects vulnerable marine species in real time
- Supports scientists with valuable ecosystem data
- Promotes cleaner, healthier oceans
- Strengthens global conservation and climate research

PRODUCT SCOPE

- The scope of OceanGuard AI is to serve as an intelligent monitoring and response system for marine conservation
- Focus: detection, alerts, deterrents, analytics
- Who benefits: biologists, conservation teams, fisheries, researchers

EXAMPLE SCENARIOS

- Biologist gets shark alert
- Conservation team protects turtles
- Fisheries adjust fishing zones

CONSTRAINTS

- **Technology: AI models, PostgreSQL, Docker/Kubernetes**
- **Environment: Harsh conditions, offline mode, sunlight-readable displays**
- **Practical: Pilot in 18 months, rollout in 3 years, $5–10M budget**

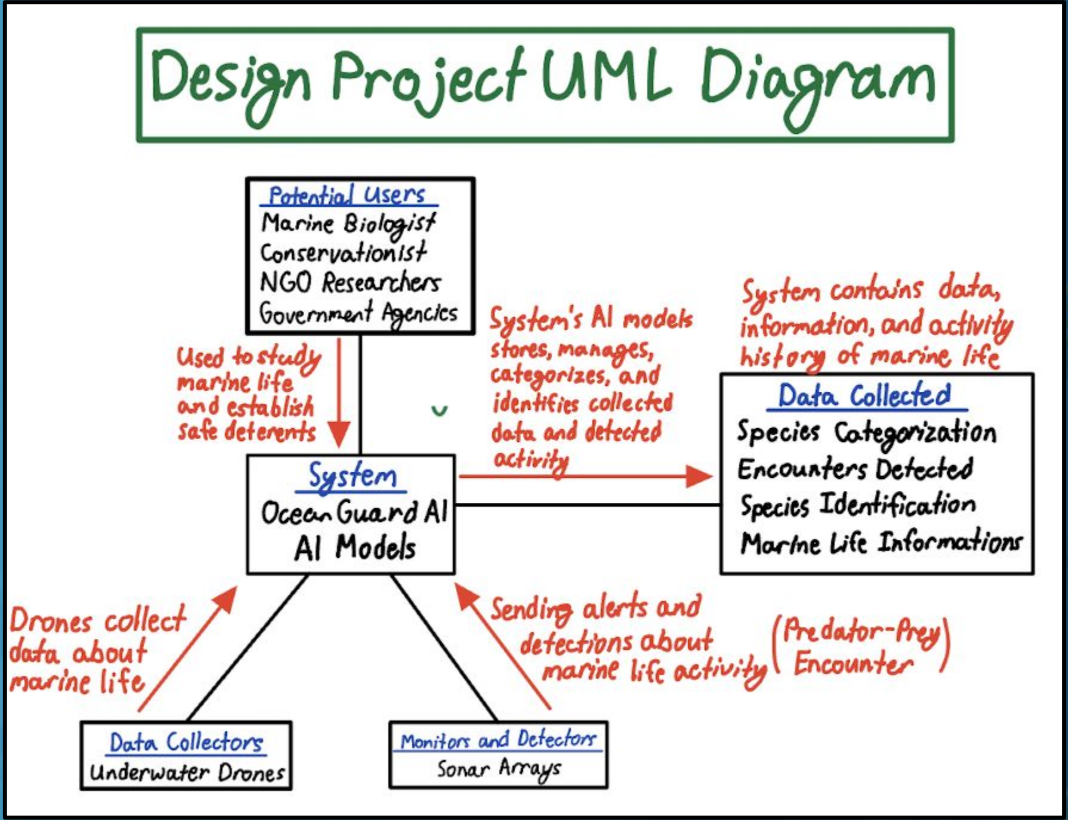
RELEVANT FACTS

- No real-time global monitoring system exists today
- Marine biologists, conservationists, and governments are actively looking for technologies protecting marine life
- Climate change and human impacts has increase biodiversity loss over the years

ASSUMPTIONS

- Under harsh ocean conditions, we expect that our underwater drones and sonar arrays can operate properly and effectively
- We hoped our AI models can capture close-to accurate and reliable data in terms of species classification and identification
- Global regulations and ethical compliance regarding environmental protection and endangered species should be satisfied

UML DIAGRAM AND DATA DICTIONARY



PLANNED DATA ENTITIES

Species

Stores and identifies marine species and their information

- SpeciesID
- Name
- Scientific Name
- Diet/Preys
- Predators
- Risk Levels

Detectors and Monitors

Sonar arrays and sensor monitoring devices network

- DeviceID
- NetworkCoverageArea
- LocationOfDevice
- DeviceSensitivity
- DeviceStatus
- DetectedEncounters
- DetectedMovements
- Description

Alerts

Critical risk level of species and dangerous encounters

- AlertID
- EncounterID
- PriortyLevel
- SpeciesThreatened
- SpeciesThreat

Encounters

Detected Encounters and Activities Between Species

- EncounterID
- SpeciesInvolved
- LocationOfEncounter
- TimeofEncounter
- TypeOfEncounter
- Description

Data Collector

Data gathering devices and underwater drones

- DeviceID
- DeviceType
- LocationOfDevice
- DeviceStatus
- DataCollected

User/Researcher

The user or researcher registered in the system

- UserID
- UserProfession
- PhoneNumber
- Email
- StartDate

CONCLUSION

Ocean Guard AI is the marine protection system, using drone, sonar, and AI.

It protects endangered marine species and supports scientific research by providing reliable data.

It contributes to a sustainable ocean ecosystem.

QUESTIONS?

THANK YOU FOR LISTENING!

REFERENCES

Visual assets (images, graphics, and videos) were sourced and adapted from Canva’s design platform.